



LncRNA and mRNA Expression Profiling in the Periurethral Vaginal Wall Tissues of Postmenopausal Women with Stress Urinary Incontinence

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Abstract

Stress urinary incontinence (SUI) is one of the major pelvic floor disorders affecting postmenopausal women. To investigate the lncRNA and mRNA expression profiling of SUI in postmenopausal women, we used a microarray analysis to examine the differentially expressed long noncoding RNAs (lncRNAs) and messenger RNAs (mRNAs) in the periurethral vaginal wall of postmenopausal women with SUI. A total of 8840 lncRNAs and 7102 mRNAs were dysregulated in the two groups (absolute fold change ≥ 2 and $P < 0.05$). The expression levels of five randomly selected lncRNAs and mRNAs were validated by quantitative real-time PCR. A functional analysis revealed that several lncRNAs are involved in the lysosome pathway associated with extracellular matrix (ECM) remodeling. In addition, we also found several mRNAs involved in fibroblast pseudopodia formation, fibroblast growth, and the regulation of smooth muscle cell differentiation in the urinary tract. Our study offers essential data regarding differentially expressed lncRNAs and mRNAs and may provide new potential candidates for the study of SUI.

Keywords Stress urinary incontinence · lncRNA · mRNA · Microarray

Introduction

Stress urinary incontinence (SUI), a common type of urinary incontinence (UI), seriously affects the health of menopausal women and reduces their quality of life [1]. The International Continence Society defines SUI as follows: a lack of detrusor contraction and an increase in the intraabdominal pressure that

results in a subsequent increase in the intravesical pressure, exceeding the maximum urethral pressure and resulting in an involuntary loss of urine, such as during coughing, sneezing, or physical exercise [2, 3]. Epidemiological studies have shown that up to 25% of women aged 45 to 64 years and 30% of those aged 65 years have symptoms of UI. Even worse, the percentage of women aged over 65 years who

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experience daily UI is 28% [4]. It is reported that the prevalence of SUI is between 10% and 58.4% and increases with an aging society [5, 6]. Although it is not a life-threatening disease, SUI can cause social withdrawal and peer rejection; it also affects physical and mental health, leads to rejection by relatives, and compromises quality of life.

The complete mechanical stability of the lower urinary system is considered critical to the mechanism of continence. However, the etiology of SUI is complex and not fully understood. Risk factors for the development of SUI include vaginal deliveries, previous pelvic trauma, race, age, and estrogen deficiency [7, 8].

It is puzzling why some nulliparous women experience SUI, while others who delivered vaginally have no symptoms [9]. On the other hand, epidemiological investigations have shown that women with UI had a family history of UI [10–12]. These observations, coupled with epidemiological data [8, 13, 14], indicate that there may be genetic differences in females that contribute to stress urinary incontinence and pelvic tissue degradation.

In recent years, long noncoding RNAs (lncRNAs), which are defined as RNA transcripts longer than 200 nucleotides with no apparent protein-coding potential [15], have attracted significant attention due to their emerging functions in development and disease [16, 17]. lncRNAs are diverse RNAs that participate in many biological processes throughout life, such as allosterically regulating enzymatic activity, influencing chromosome conformation, and imprinting genomic loci [18, 19]. According to their genomic proximity to protein-coding genes, lncRNAs can be classified into six categories: intronic antisense, intron sense-overlapping, natural antisense, exon sense-overlapping, bidirectional, and intergenic [19]. A number of lncRNAs have been confirmed to have important functions in recruiting regulatory complexes via RNA-protein interactions to regulate the expression of nearby genes [20, 21], which suggests that many other lncRNAs can also act as local regulators [22, 23]. In most cases, the evidence depends on the differences in transcript expression levels between disease and non-disease-related states [24]. Therefore, the lncRNA and mRNA expression profiles of SUI will help us better understand the pathogenesis of SUI.

However, few studies have focused on the potential genetic predisposition to SUI, and no lncRNA expression profiles of SUI have been reported until now. As lncRNAs have been implicated its potential function in the development of diseases, we hypothesized that lncRNAs may contribute to the molecular mechanisms of SUI. We also performed the transcriptomic analysis of mRNA in vaginal samples of women with and without SUI. In our study, we identified the differentially expressed lncRNAs in the periurethral vaginal wall tissue of postmenopausal women with and without SUI via high-throughput microarray analyses. We believe that this study will not only help researchers better understand the

molecular pathogenesis of SUI but also provide further insight into SUI therapy.

Materials and Methods

Patient Selection

This study was approved by the Research Ethics Committee of Nanjing Maternity and Child Health Care Hospital Affiliated with Nanjing Medical University (Nanjing, China). To minimize the effect of reproductive hormones on periurethral vaginal wall tissue metabolism, we selected postmenopausal women who did not receive hormone therapy as study participants. This study group involved 22 participants (Control vs SUI, 11 vs 11) who were enrolled at Nanjing Maternity and Child Health Care Hospital Affiliated with Nanjing Medical University. Women with a history of a prior pelvic surgical operation, connective tissue disorders, gynecologic malignancies, endometriosis, and advanced pelvic organ prolapse (higher than stage II on the Pelvic Organ Prolapse Quantification System) were excluded. All of the patients with urinary incontinence symptoms in this study were diagnosed via a comprehensive gynecological examination, including the pelvic organ prolapse quantification (POPQ) test, a urodynamic examination, ultrasonography, a urinary stress test, and a gynecological examination. All controls have no stress urinary incontinence but with benign uterine fibroids by comprehensive gynecological examination from the same hospital.

Tissue Collection

Informed consent was obtained from all subjects. Approximately 1 cm² of full-thickness periurethral vaginal wall was excised 1 cm lateral to the ureterovesical junction from postmenopausal women undergoing a surgical operation for SUI. Similarly, 1 cm² biopsies of the vaginal wall from patients asymptomatic for SUI, but undergoing a transvaginal gynecologic surgical operation, were designated as "controls". All the tissues were frozen in liquid nitrogen immediately after removal from women and stored at − 80 °C. Of these samples, six samples were used for the lncRNA and mRNA microarray analysis, and the other samples were assessed by quantitative reverse transcription-polymerase chain reaction (qRT-PCR) for further validation.

RNA Extraction and Quality Control

Total RNA was harvested from the frozen periurethral vaginal wall tissue using TRIzol reagent (Invitrogen, Carlsbad, CA, USA) according to the protocol suggested by the

manufacturer. The quantification and quality of the extracted RNA were determined on a NanoDrop ND-2000 (Thermo Scientific Co., USA). For all the samples, the O.D. A260/A280 ratio was between 1.8 and 2.1. RNA integrity and the gDNA contamination test were assessed by standard denaturing agarose gel electrophoresis.

Microarray Analysis

The microarray analysis was performed using an Arraystar Human lncRNA and messenger RNA (mRNA) Microarray V4.0 (8 × 60, format). Approximately 40,173 lncRNAs and 20,730 coding genes were collected from the most authoritative public transcriptome databases, such as UCSC Known Genes and Gencode (genome.ucsc.edu/index.html), Gencode (www.gencodegenes.org/), and RefSeq (NCBI Reference Sequence Database; www.ncbi.nlm.nih.gov/refseq/), and landmark publications [22, 25–27] can be detected via this fourth-generation lncRNA microarray. Sample labeling and array hybridization were performed using the Agilent One-Color Microarray-Based Gene Expression Analysis according to the manufacturer's standard protocol (Agilent Technology). In brief, mRNA was purified from total RNA after the removal of rRNA using an mRNA-ONLY™ Eukaryotic mRNA Isolation Kit (Epicentre, Madison, WI, USA). Then, each sample was amplified and transcribed into a fluorescently labeled cRNA along the entire length of the transcript without 3' bias utilizing a mixture of oligo (dT) and the random priming method (Flash RNA Labeling Kit; Arraystar, Rockville, MD, USA). The labeled cRNAs were purified via an RNeasy Mini Kit (Qiagen, Inc., Valencia, CA, USA). The concentration and specific activity of the labeled cRNAs (pmol Cy3/μg cRNA) were measured on a NanoDrop ND-1000. Fifty microliters of hybridization solution was dispensed into the gasket slide and then assembled with the lncRNA expression microarray slide. The slides were incubated for 17 h at 65 °C in an Agilent hybridization oven. The hybridized arrays were washed, fixed, and scanned using an Agilent DNA Microarray Scanner (part number G2505C). Agilent Feature Extraction software (version 11.0.1.1, Agilent Technologies) was used to analyze the acquired array images. Raw data were normalized by a quantile algorithm using the GeneSpring GX v12.1 software package (Agilent Technologies). Differentially expressed mRNAs and lncRNAs were defined by an absolute value of fold change (FC) ≥ 2 and $P < 0.05$. Throughout the computational analysis, the false discovery rate (FDR) was calculated using the Benjamini-Hochberg method. Hierarchical clustering was performed based on the differentially expressed mRNAs and lncRNAs using Cluster-TreeView software from Stanford University. In our study, we

compared three samples of the periurethral vaginal wall with SUI to matched tissues without SUI using the Arraystar microarray analysis.

qRT-PCR

Total RNAs isolated from frozen periurethral vaginal wall tissues were reverse-transcribed to cDNA using a PrimeScript™ RT Reagent Kit (TaKaRa Bio, Inc., Japan) in accordance with the manufacturer's recommendations. The expression of lncRNAs and mRNAs in periurethral vaginal wall tissues was measured by qRT-PCR using SYBR Premix Ex Taq (TaKaRa Bio, Inc., Japan) on an ABI 7900HT instrument (Applied Biosystems). Glyceraldehyde-3-phosphate dehydrogenase (GAPDH) was used as an internal control. All primer sequences used in this study are listed in Table 1. Briefly, the PCR conditions were 95 °C for 30 s, followed by 40 PCR cycles of 95 °C for 5 s and 60 °C for 30 s. Finally, annealing and extension at 95 °C for 15 s, 60 °C for 60 s, and 95 °C for 15 s were performed. We calculated the relative expression of each gene with the formula $2^{-(\Delta\Delta C_t)}$ [28].

Gene Ontology and Pathway Analyses

Gene ontology (GO) and pathway analyses were applied to identify the potential functions of the differentially expressed genes in GO terms and biological pathways. The Gene Ontology project was used to describe genes and gene product attributes (<http://www.geneontology.org>), including molecular functions, cellular components, and biological processes. The differentially expressed predictive target genes of the lncRNAs were mapped to GO terms in the Database for Annotation, Visualization, and Integrated Discovery (DAVID) (<http://david.abcc.ncifcrf.gov/>), which was used to select the main pathways. Fisher's exact test was used to determine whether there was more overlap between the DE list and the GO annotation list than occasionally expected. The P value indicates the significance of the GO term enrichment in the list of differentially expressed genes (a P value ≤ 0.05 is recommended).

The pathway analysis based on the functional analysis of the Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways (www.genome.jp/kegg/). This analysis was performed to locate the significant enrichment pathways and to further understand the potential biology of the differentially expressed genes. The P value represents the significance of the conditional-related pathway. The lower the P value, the more significant the pathway is (a P value ≤ 0.05 is recommended).

Table 1 Gene-specific primer sequences for real-time quantitative PCR

Gene	Forward sequence	Reverse sequence
GAPDH	CTGGGCTACACTGAGCACC	AAGTGGTCGTTGAGGGCAATG
CSTA	AGGCTTATCTGAGGCCAAACC	GCACAGCTTCCAATTTTCCG
CRISP3	TGCTCTGGAACCACTGCAA	GCATTTGCTGCAGCCTCTTT
CNMM1	AGCTTTCGGACACGGAGATG	CGGGGCCTTCTTGTTCTTCT
CDC42EP4	CCTTTGGGGACACCTCCTTC	TCTTGGGCAGCTTACTGGTG
HTR5A	ATCCTCATTGGCGTGTTCTG	AGCCGTATAGATCAGGGGGT
C17orf76-AS1	ACTACAGCTCACAGCACACCT	TGGCAAGGGACAAAGCTACAA
XLOC-000669	CATGACACCACAAACCACCCC	GCCAGAAAGATAGGGGTGTCCG
RP11-631B21.1	CAGGGCTCCAAAGGCTCGTTT	TCTGGGCCTTTGTGTTTGTCT
LOC100499194	CGCCTCAAACGTAAACGGC	ACCAACTGGCTTTTGCGAAC
XLOC_006298	CTGAGCTACGCGATTCTCCC	CAGGCTTTTCTCATTTCTCAGGT

Statistical Analysis

Student's *t* test was used to evaluate the differential expression levels of the lncRNAs and mRNAs between the SUI and control groups, and a fold change ≥ 2.0 and $P < 0.05$ were considered significant. The false discovery rate (FDR) was calculated to correct the *P* value. For GO and KEGG pathway analyses, Fisher's exact test was used to evaluate the significance of the GO terms or pathway identifier enrichment in the differentially expressed genes. $P < 0.05$ was considered to indicate a statistically significant difference. Statistical analysis was performed with SPSS version 20.0 (SPSS, Inc., Chicago, IL, USA).

Results

Participant Characteristics

To avoid the influence of confounding factors, we analyzed age, parity, and mean body mass index between the two patients' groups. No significant differences were detected in these confounding factors (Table 2). H&E staining was performed to examine periurethral vaginal wall tissues from postmenopausal women with and without SUI (SUI vs. Ctrl). H&E staining showed obvious fibrosis, intravascular congestion, and capillary proliferation in the SUI periurethral vaginal wall (Fig. 1).

Table 2 Clinical characteristics of women with and without SUI

Characteristics	SUI (<i>n</i> = 11)	Control (<i>n</i> = 11)	<i>P</i> value
Age (years)	59.30 $\pm$ 7.40	55.70 $\pm$ 6.80	0.59
Parity (median)	3.00 $\pm$ 1.10	2.40 $\pm$ 0.80	0.37
Mean body mass index (kg/m ²)	27.00 $\pm$ 3.80	24.30 $\pm$ 3.80	0.45

Differentially Expressed lncRNAs and mRNAs

Our study examined the lncRNA and mRNA expression profiles in periurethral vaginal wall tissues from postmenopausal women with and without SUI (SUI vs. Ctrl) using a microarray analysis. Hierarchical clustering analysis was applied to detect the differentially expressed lncRNAs and mRNAs in the SUI and control groups based on their expression levels (Fig. 2a–d). The volcano plot generated from the significant differentially expressed lncRNAs and mRNAs on three pairs of periurethral vaginal wall tissues showed distinct expression patterns of these lncRNAs and mRNAs between the SUI and control groups (Fig. 2e and f). In total, the expression profiles of 8840 lncRNAs (Supplementary Table S1) and 7102 mRNAs (Supplementary Table S2) were identified in the SUI and control groups (absolute fold change ≥ 2 and $P < 0.05$). Among them, 4968 lncRNAs were upregulated, and 3872 lncRNAs were downregulated, while 4615 mRNAs were upregulated, and 2487 mRNAs were downregulated. The expression profiling data revealed the most significant differentially expressed lncRNAs between the SUI and control groups, including the top 20 upregulated lncRNAs (Table 3) and the top 20 downregulated lncRNAs (Table 4). Tables 5 and 6 showed the twenty most significant differentially expressed mRNAs. We also analyzed the functions of the top 20 dysregulated mRNAs with GeneCards (<https://www.genecards.org/>), Mouse Genome Informatics (<http://www.informatics.jax.org/>), and UniProtKB/Swiss-Prot (<https://www.uniprot.org/>) (Tables 5 and 6).

qRT-PCR Validation

To verify the dysregulated expression of mRNAs and lncRNAs found by microarray, we randomly selected five mRNAs and lncRNAs among the dysregulated expression of top 10 mRNA and lncRNA transcripts. qRT-PCR was performed in 8 vs 8 tissue samples to verify the differential

expression of selected lncRNA and mRNA. The differences in the expression levels of the five mRNAs and lncRNAs between the SUI and control groups were determined (Fig. 3). The expression trends of five mRNAs and three lncRNAs were consistent with the microarray data.

lncRNA Classification and Subgroup Analysis

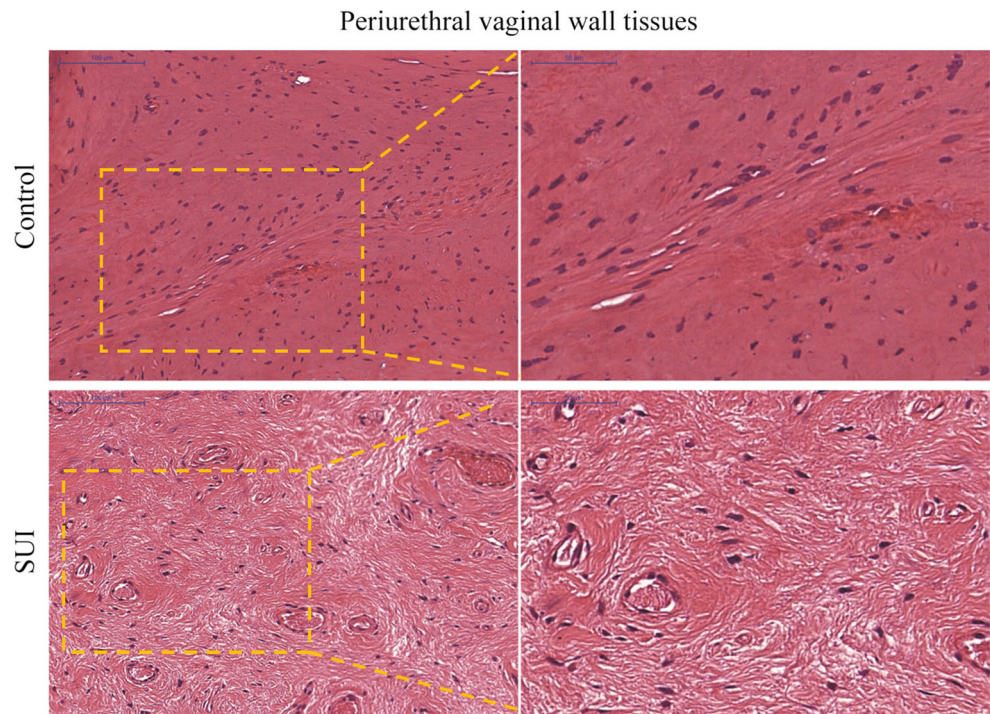
Figure 4 showed the lncRNA classification and distribution of chromosomes based on the general signatures of the dysregulated lncRNAs. The dysregulated lncRNAs were divided into six categories based on the relative positions of the lncRNAs and their nearby coding genes. These six categories included exon sense-overlapping, intronic antisense, natural antisense, intron sense-overlapping, and intergenic bidirectional. In our microarray data, the majority of lncRNAs were intergenic lncRNAs. According to the lncRNA expression profiles, we found 3417 differentially expressed intergenic lncRNAs (fold change ≥ 2 , $P < 0.05$) between the SUI group and the control group (Supplementary Table S3). In addition, there were three subgroups of lncRNAs in our microarray data: antisense lncRNAs, enhancer lncRNAs, and long intergenic noncoding RNAs (lncRNAs). In the study of lncRNAs, most researchers pay attention to antisense lncRNAs, which are reported to regulate the expression of genes in three processes of transcription (pretranscriptional, transcriptional, and posttranscriptional) via DNA-lncRNA, RNA-lncRNA, or protein-lncRNA interactions [29]. In the antisense lncRNA subgroup,

476 antisense lncRNAs were differentially expressed between the SUI group and the control group (fold change ≥ 2 , $P < 0.05$). Among these, 245 were upregulated, and 231 were downregulated (Supplementary Table S4).

Gene Ontology and Pathway Analyses

To further analyze the potential functions and mechanisms of these upregulated and downregulated lncRNAs in postmenopausal women paired with periurethral vaginal wall tissues (SUI vs. Ctrl), we applied GO and pathway analyses of lncRNAs with the associated differentially expressed mRNAs (< 100 kb). The GO analysis was carried out to describe each gene and gene product enrichment in cellular components, biological processes, and molecular functions. In the GO analysis, we identified the top ten significant GO terms of three domains based on the enrichment score ($-\log_{10}(P \text{ value})$) of the dysregulated genes. Regarding biological processes, protein localization to the endoplasmic reticulum (GO: 0070972) was the most enriched term among the upregulated transcripts (Fig. 5a), while protein localization to the positive regulation of biological processes (GO: 0048518) was the most enriched term among the downregulated transcripts (Fig. 5b). Regarding cellular components, the cytoplasm (GO: 0005737) was the most enriched term among the upregulated transcripts (Fig. 5a). In contrast, the downregulated genes were mostly enriched in the term nucleus (GO: 0005634) (Fig. 5b). Regarding molecular functions, the

Fig. 1 Hematoxylin and eosin (H&E) staining of periurethral vaginal wall tissues from the control and SUI groups. Scale bars, 100 μm and 50 μm



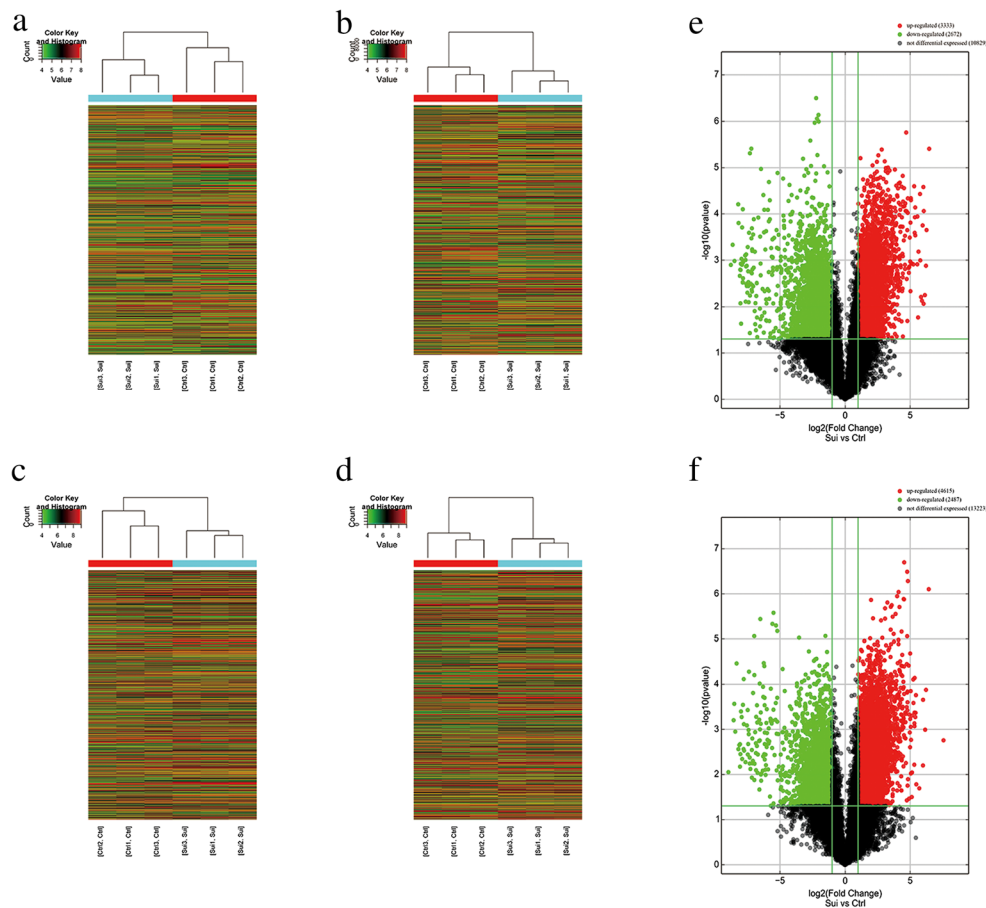


Fig. 2 Differentially expressed lncRNAs and mRNAs in SUI. **a, b** Transcriptome hierarchical cluster analysis revealed 16,834 lncRNAs among the control and SUI groups, 6005 of which were significantly differentially expressed (3333 lncRNAs were upregulated (red), and 2672 lncRNAs were downregulated (green), with absolute fold changes ≥ 2 and $P < 0.05$). **c, d** Transcriptome heatmap showing the clustering of 20,325 mRNAs expressed among the control and SUI groups, 7102 of which were differentially expressed (4615 mRNAs were upregulated

(red), and 2487 mRNAs were downregulated (green), absolute fold change ≥ 2 and $P < 0.05$). **e** The volcano map of the differentially expressed lncRNAs. **f** The volcano map of the differentially expressed mRNAs. X-axis, log2 represents the fold change; Y-axis, expressed as the p value. The two vertical green lines represent 2-fold upregulation and downregulation, and the horizontal green line represents a P value of 0.05. The red and green dots represent the lncRNAs and genes that are significantly upregulated and downregulated, respectively

upregulated transcripts were mostly enriched in the structural constituent of ribosomes (GO: 0003735) (Fig. 5a), and the downregulated genes were mostly enriched in protein binding (GO: 0005515) (Fig. 5b).

Pathway analyses indicated 25 pathways that corresponded to the upregulated mRNA transcripts and that the ribosome pathway was the most enriched among the upregulated pathways, which included 58 differentially expressed mRNA transcripts (Fig. 5c). Moreover, the GO analysis also showed that 58 pathways corresponded to the downregulated mRNA transcripts. Among the downregulated pathways, the regulation of the actin cytoskeleton pathway was the most enriched and contained 45 differentially expressed mRNA transcripts. The other main pathways included transcriptional misregulation in cancer, fc gamma R-mediated phagocytosis, the lysosome, Alzheimer's disease, and the Parkinson's disease signaling pathway (Fig. 5d).

Discussion

The pathogenesis of SUI is complex and not well understood. Previous studies have demonstrated that vaginal delivery, increased age, smoking, and obesity were risk factors for SUI [30]. However, several studies have confirmed that genetic factors are also involved in the underlying pathogenesis of SUI [31–33]. Epidemiological investigations have illustrated that women with urinary incontinence usually have a family history of UI [11, 34, 35], and genetic factors may account for approximately 40% of the variation in SUI and pelvic organ prolapse [36]. The periurethral vaginal wall is the main supporting tissue and provides a stable base to prevent urethral and bladder neck descent [37]. The vaginal wall contains squamous epithelium, subepithelium, smooth muscle, and adventitia, all of which are embedded in an extracellular matrix extracellular matrix (ECM) composed of laminin, collagen, and elastin fibers [38]. In periurethral support tissues, collagen

Table 3 Top 20 upregulated expressed lncRNAs

Seqname	Gene name	Fold change	RNA type
ENST00000460249	C17orf76-AS1	88.0930826	ncRNA
ENST00000503267	RP11-321E2.4	76.6031776	ncRNA
ENST00000475362	RP11-631B21.1	70.0274412	ncRNA
ENST00000567829	LA16c-358B7.3	66.1131783	ncRNA
ENST00000549364	RP11-804F13.1	65.4617954	ncRNA
ENST00000443776	RP11-407B7.1	64.6756421	ncRNA
ENST00000440438	AC005154.9	62.1841071	ncRNA
TCONS_00017996	XLOC_008852	57.1709332	ncRNA
TCONS_00001368	XLOC_000669	54.7663641	ncRNA
ENST00000441570	LINC00472	54.0498284	ncRNA
ENST00000503167	RP11-168A11.1	49.9398241	ncRNA
NR_073068	BCL2L13	48.8156285	ncRNA
ENST00000434273	ENOX1-AS2	47.1300639	ncRNA
ENST00000472293	C17orf76-AS1	46.8443658	ncRNA
ENST00000562852	RP11-143N13.2	42.0360683	ncRNA
ENST00000458001	GAS6-AS1	41.8479429	ncRNA
TCONS_00019479	XLOC_009282	41.1343042	ncRNA
ENST00000510844	RP11-33B1.1	40.7206165	ncRNA
TCONS_00008034	XLOC_003498	39.6176024	ncRNA
ENST00000447563	AC004863.6	37.9955626	ncRNA

types I and III are the main structural components of the ECM. In women with SUI, decreases in collagen levels in the pelvic fascia and anterior vaginal wall are implicated in abnormal

collagen metabolism in the pathogenesis of SUI [39, 40]. Although the strength of connective tissue is derived from collagen, elasticity and recoil ability are supported by elastin. The remodeling of collagen and elastin when elastin fibers are damaged may then result in weak fibers prone to failure [41, 42]. Vaginal tissue tensile strength and its mechanical stability are largely determined by the ECM of the vaginal tissue. Several studies also showed that the genetic factors mainly affect remodeling of the ECM and collagen synthesis [33, 43].

In this study, we identified 7102 dysregulated mRNAs, including 4615 upregulated genes and 2487 downregulated genes. To further explore the potential functions of these genes, we analyzed the top 20 dysregulated genes with the GeneCards, Mouse Genome Informatics, and UniProtKB/Swiss-Prot databases. Our functional analysis revealed that these dysregulated genes are involved in the regulation of smooth muscle cell differentiation in the urinary tract, fibroblast pseudopodia formation [44], and fibroblast growth (Tables 5 and 6). Fibroblasts are the main cells that respond to mechanical forces and have the function of synthesizing and secreting the ECM. However, mechanical force plays an important role in the pathogenesis of female SUI, and pregnancy, race, chronic disease, and fetal size resulting in chronic and acute mechanical damage are risk factors for female SUI [45]. Therefore, these dysregulated genes may play an important role in female SUI.

Previous studies have demonstrated that miRNA and mRNA play important roles in the pathogenicity of SUI [33,

Table 4 Top 20 downregulated expressed lncRNAs

Seqname	Gene name	Fold change	RNA type
NR_034130	LOC100499194	439.3716579	ncRNA
uc002thg.1	AX747531	398.8302608	ncRNA
ENST00000566944	FAM157C	361.4845224	ncRNA
ENST00000431145	ZFY-AS1	296.5540505	ncRNA
NR_047583	IL12RB2	290.9850991	ncRNA
TCONS_00013636	XLOC_006298	273.0026667	ncRNA
ENST00000423727	AC009505.4	266.3882552	ncRNA
ENST00000449012	RP4-781K5.4	263.8236156	ncRNA
uc021qut.1	DDX12P	261.9811317	ncRNA
ENST00000453278	RP11-807H17.1	258.6082757	ncRNA
TCONS_00020656	XLOC_009955	254.8115742	ncRNA
NR_027779	TTL1	248.9866246	ncRNA
ENST00000517735	RP11-597M17.1	242.7903437	ncRNA
chr13:52409200-52419600	chr13:52409200-52419600	241.1181206	ncRNA
ENST00000506014	LINC00461	239.5552188	ncRNA
TCONS_00028254	XLOC_013596	233.2595492	ncRNA
ENST00000506807	AC063976.6	232.3125741	ncRNA
TCONS_00008978	XLOC_003832	230.2495192	ncRNA
ENST00000415471	RP11-100G15.10	229.8430289	ncRNA
ENST00000520924	TPT1-AS1	204.9493305	ncRNA

Table 5 Top 20 upregulated expressed mRNAs

Seqname	Gene name	Fold change	Function/disease
NM_005213	CSTA	189.111825	Acral peeling skin syndrome (PMID: 23534700)
NM_000557	GDF5	86.0918979	Brachydactyly, Type C (PMID: 18283415)
NM_020844	KIAA1456	74.4839735	tRNA processing and gene expression (GeneCards)
NM_006061	CRISP3	71.1619612	Male and female reproductive tract (PMID: 15867000, GeneCards)
NM_000983	RPL22	63.6053597	RNA binding and structural constituent of ribosome (PMID: 12962325, 22658674)
NM_001167911	VEPH1	63.3327199	Carotid Artery Diseases (PMID: 17903303)
NM_007084	SOX21	52.0067159	Colon carcinoma and colon adenocarcinoma (PMID: 23856246)
NM_016190	CRNN	44.3793921	Epidermal differentiation (PMID: 19133922)
NM_153187	SLC22A1	43.4985026	Chronic myeloid leukemia (PMID: 18559609)
NM_199259	TPTE	42.6577589	Signal transduction and alternative splicing (PMID: 14659893 and GeneCards)
ENST00000225430	RPL19	42.1089841	RNA binding and structural constituent of ribosome (PMID: 1577483)
NM_001005855	ATP8B2	40.7120802	ATP binding and phospholipid-translocating ATPase activity (GeneCards)
NM_004183	BEST1	40.4076515	Chloride channel activity (PMID: 17003041)
NM_001097599	SLC35G2	40.3027814	May play a role in cell proliferation (UniProtKB/Swiss-Prot)
NM_001029881	CIB4	40.1301942	Magnesium ion binding and calcium ion binding (PMID: 22779914)
NM_004925	AQP3	39.8809314	Water channel activity and glycerol channel activity (PMID: 17943189)
ENST00000391796	CD33	38.1263857	Signaling receptor activity (PMID: 10611343)
NM_033296	MRFAP1	37.6879417	Senescence, cell growth and protein binding (GeneCard and PMID: 11500496)
NM_020856	TSHZ3	35.2958926	Regulates smooth muscle cell differentiation in urinary tract (GeneCards)
NM_175054	HIST4H4	35.2031755	Core component of nucleosome (UniProtKB/Swiss-Prot)

46, 47]. However, the human genome has relatively few coding sequences, and the majority is composed of noncoding RNAs (ncRNAs). lncRNAs form a large class of ncRNAs and participate in numerous biological processes [48]. To study the important roles of lncRNAs in SUI, we analyzed the transcriptome of periurethral vaginal wall tissues from postmenopausal women with and without stress urinary incontinence with a microarray. We obtained a total of 8840 lncRNAs in the SUI and control groups. After comparing the SUI and control groups, of these 8840 lncRNAs, 4968 were significantly upregulated, and 3872 were significantly downregulated.

Using these data, we performed a GO analysis and found that the main biological processes involved included metabolic processes, cellular metabolic processes, biological regulation, and regulation of the cellular process. Collagen is the main structural component of the vaginal wall [49]. The balance between collagen synthesis and degradation plays an important role in maintaining tissue

integrity and tensile strength. Several studies have revealed that collagen synthesis is decreased if connective tissue metabolism is affected, which would result in insufficient support of the urogenital tract in women [50, 51]. Given this biological process, it is likely that some lncRNAs may alter connective tissue metabolism through the regulation of target gene expression and cause stress urinary incontinence in women.

Our KEGG signaling pathway analysis demonstrated that the significantly dysregulated lncRNAs are mainly involved in the somal cysteine proteinase (i.e., the lysosome), the ribosome, and regulation of the cytoskeletal pathway. Lysosomes are classically viewed as degradative organelles [52] that contain multiple proteolytic enzymes. Cathepsin B, a lysosomal cysteine proteinase, exhibits collagenolytic activity [53]. Cathepsin B is involved in ECM remodeling by increasing the activity of matrix metalloproteinase 9 (MMP-9) [54]. The ECM of the vaginal tissue largely determines tissue tensile strength and

Table 6 Top 20 downregulated expressed mRNAs

Seqname	Gene name	Fold change	Function/disease
ENST00000356713	CNNM1	500.8657323	Protein binding (PMID: 28330616)
NM_012121	CDC42EP4	379.1452487	Induce fibroblasts pseudopodia formation (GeneCards, PMID: 11035016)
NM_024645	ZMAT4	365.9546499	Protein binding and Blood pressure (PMID: 25416956, 17903303)
ENST00000262738	CELSR1	347.2495144	Transmembrane signaling receptor activity (PMID: 9339365)
NM_017929	PEX26	323.3320552	Protein-containing complex binding, ATPase binding (PMID: 16854980, 16257970)
NM_024012	HTR5A	293.839289	G protein-coupled receptor activity (PMID: 21873635)
NM_199132	ZNF468	280.1376988	Transcriptional regulation nucleic acid binding (UniProtKB/Swiss-Prot, GeneCard)
NM_004561	OVOL1	270.0566006	Hair formation and spermatogenesis (GeneCards, PMID: 25509883, 9808631)
NM_001135731	ZMAT4	269.4769813	Protein binding and zinc ion binding (PMID: 25416956 and GeneCards)
NM_014732	KIAA0513	263.9823079	Neuroplasticity, apoptosis, and the cytoskeleton (PMID: 17010949)
NM_020225	STOX2	256.2057422	The susceptibility to pre-eclampsia with fetal growth restriction (GeneCards)
ENST00000394168	KIAA0240	246.6267253	Chromatin remodeling subcomplexes (PMID: 29374058)
ENST00000449195	IL4R	240.433009	Abnormal cutaneous collagen fibril morphology (Mouse Genome Informatics)
NM_003867	FGF17	230.4300141	Fibroblast growth factor and (PMID: 16384934 and GeneCards)
NM_001760	CCND3	222.8223338	Decreased B cell proliferation and protein kinase binding (Mouse Genome Informatics)
NM_182595	POM121L12	220.9735425	Structural constituent of nuclear pore (PMID: 21873635)
NM_017650	PPP1R9A	219.700661	Neurite formation (UniProtKB/Swiss-Prot and Mouse Genome Informatics)
NM_001134647	AFAP1	217.1602039	Actin filament integrity and induce lamellipodia formation (PMID: 14755689)
ENST00000356175	NF1	201.049727	Neurofibromatosis-Related Protein, GTPase activator activity (PMID: 1568247)
NM_001159495	BANF2	186.863938	Testis Secretory Sperm-Binding Protein (GeneCards and PMID: 21630459)

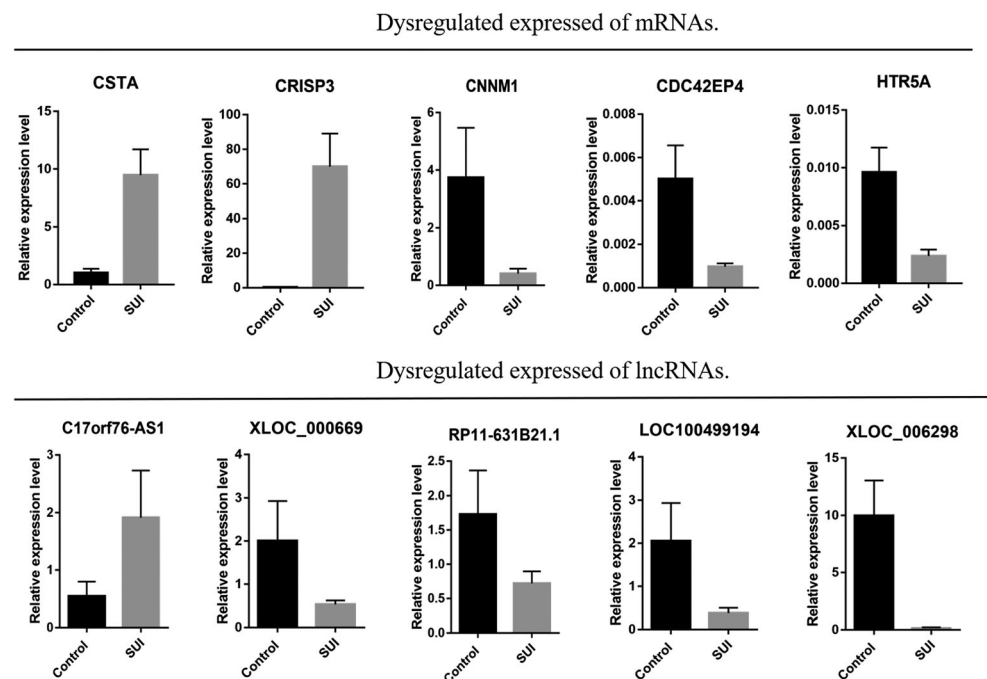
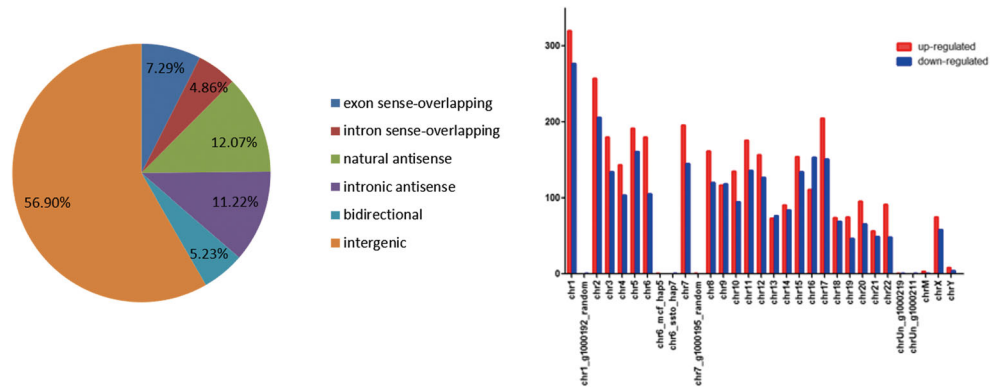
Fig. 3 qRT-PCR results of five randomly selected mRNAs and lncRNAs, respectively. The verified qRT-PCR results of five mRNAs and three lncRNAs were consistent with the microarray data

Fig. 4 lncRNA classification and the distribution of chromosomes. **a** The dysregulated lncRNAs were divided into six categories. **b** The distribution of chromosomes based on the general signatures of the dysregulated lncRNAs



mechanical stability. These findings indicate a potential role for lncRNAs in women with SUI.

In this study, we focused on the role of lncRNAs in SUI. As mentioned previously, many lncRNAs play a critical role in disease occurrence. However, work regarding the role of lncRNAs in women with SUI is lacking. This is the first study

to show that the differentially expressed lncRNAs are associated with the etiology of SUI in women. We hope that the results will be beneficial for studying the functions and molecular mechanisms of lncRNAs and mRNAs in women with SUI, as well as applied to genetic screening for women at risk of SUI and available for SUI treatment in the future.

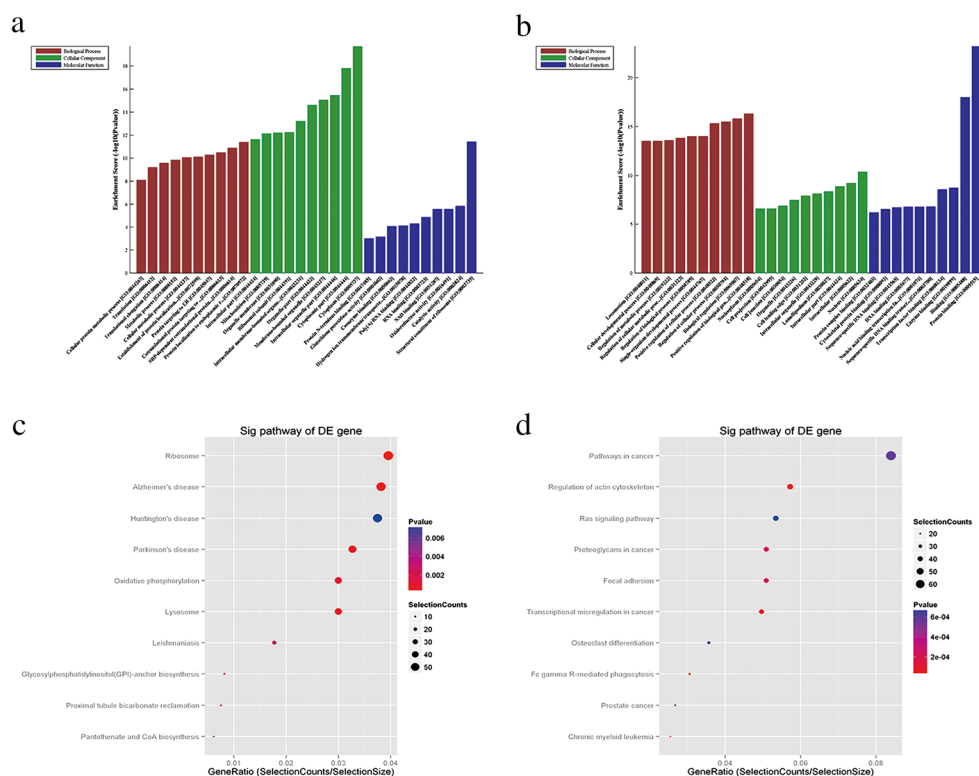


Fig. 5 GO and pathway analyses of the lncRNAs and their associated differentially expressed mRNAs. The most significantly enriched GO terms for the upregulated (**a**) and downregulated (**b**) transcripts are displayed. The GO terms for molecular function are shown in blue; those for cellular component are shown in green; and those for biological process are shown in red. Bubble map of the pathway

analysis of (**c**) the upregulated mRNA-related significantly enriched signaling pathways and (**d**) the downregulated mRNA-related significantly enriched signaling pathways. The Y-axis shows the name of the enriched pathway. The X-axis shows the gene ratio (selection counts/selection size). The size of the dot in the bubble map indicates the number of enriched mRNAs, and the red indicates the *P* value

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Author Contributions YF and YZ directed the study and were responsible for study design. AMW and RHW performed overall project management with KW. CCD and YH performed data analysis with PFX and drafted the initial manuscript. JX and HT were responsible for subject recruitment and sample preparation. All authors read and approved the final manuscript.

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Compliance with Ethical Standards

Conflict of Interests The authors declare that they have no competing interests.

Research Ethics This study was approved by the Research Ethics Committee of Nanjing Maternity and Child Health Care Hospital Affiliated with Nanjing Medical University (Nanjing, China).

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